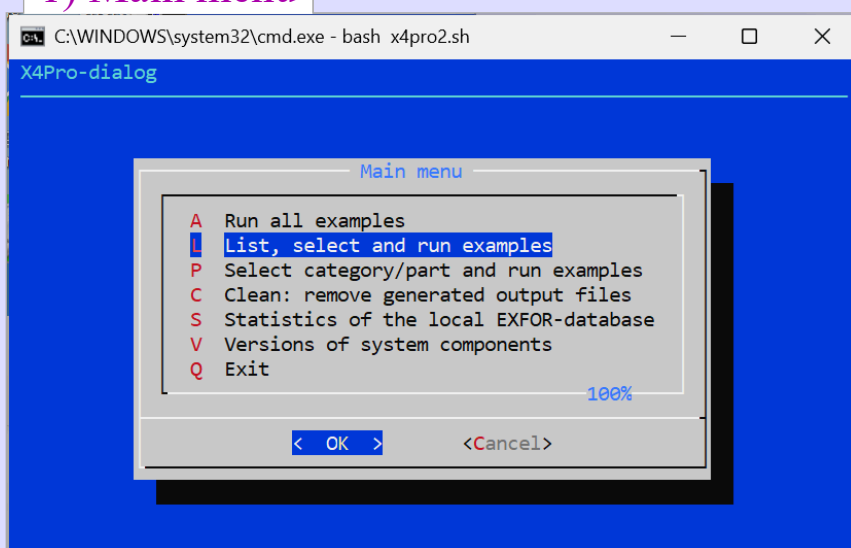


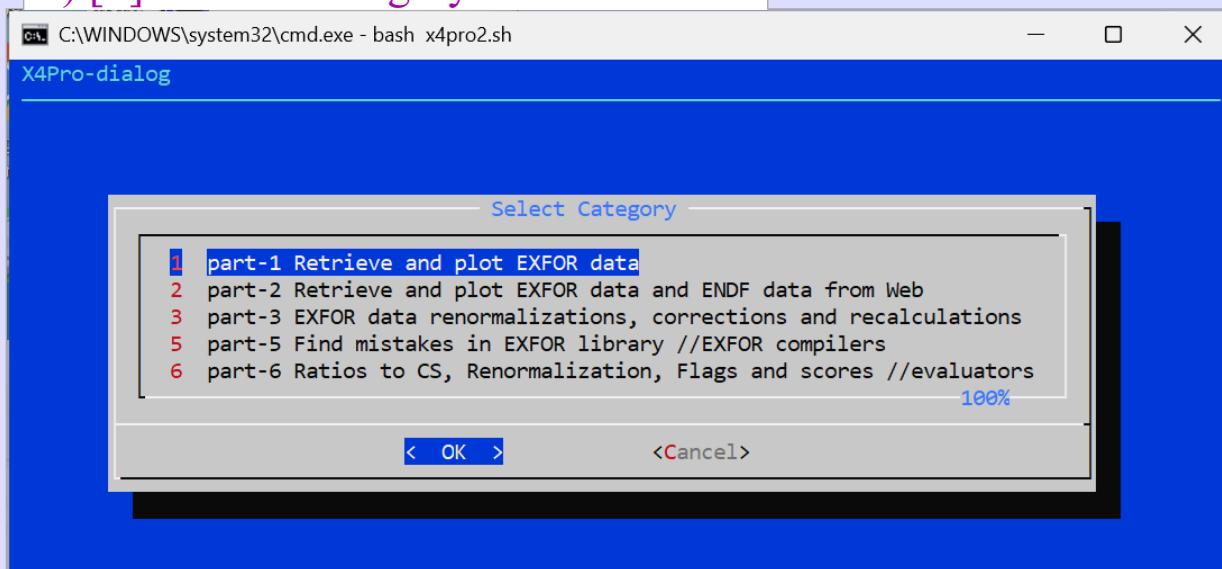
Menu-type interactive script **x4pro2.sh** using 'dialog' utility

```
# Script: x4pro2.sh
# Author: Viktor Zerkín
# Created: 2026-02-19
# Version: 2026-02-23
# License: MIT
# Run: $ [bash] x4pro2.sh
# Example: $ bash x4pro2.sh
# $ ./x4pro2.sh
# Main menu:
# A Run all examples
# L List, select and run examples
# P Select category/part and run examples
# C Clean: remove generated output files
# S Statistics of the local EXFOR-database
# V Versions of system components
# Q Exit
```

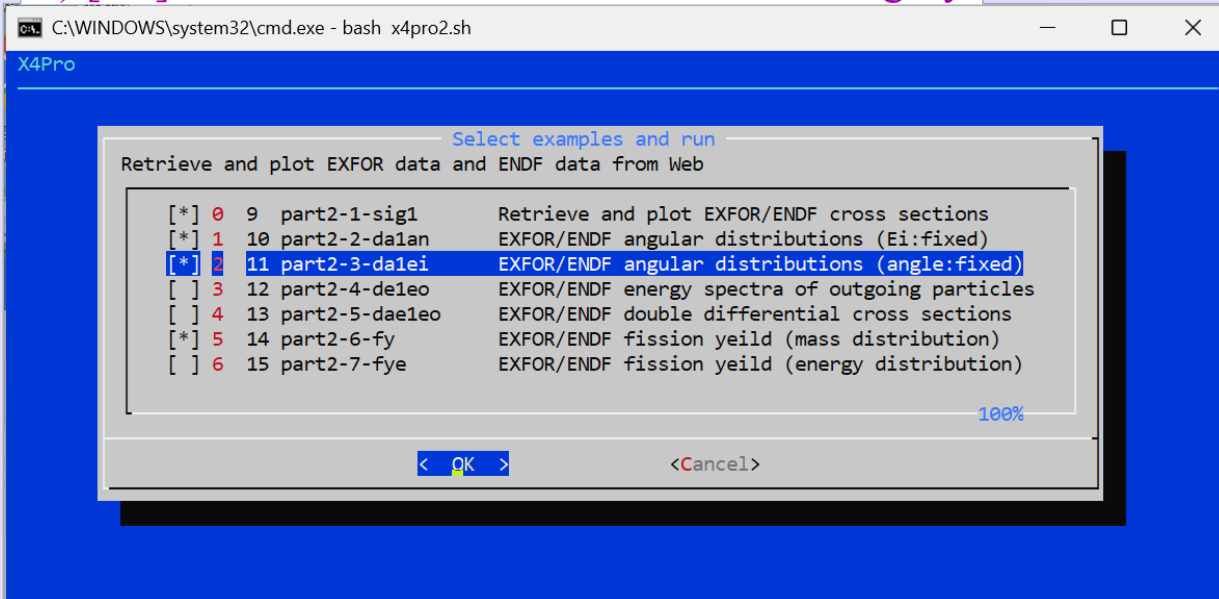
1) Main menu



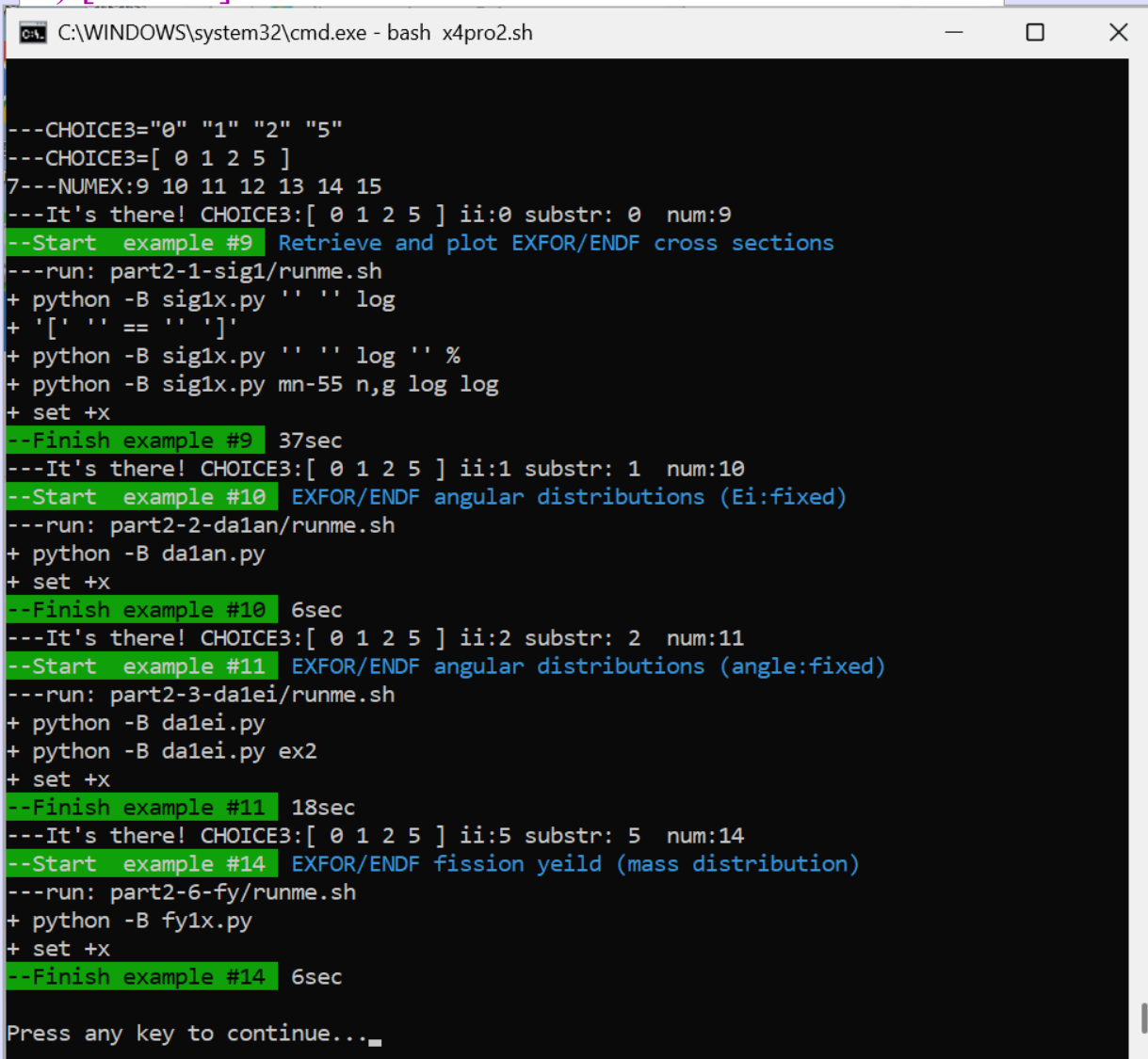
2) [P] - Select Category of demo-codes



3) [P.2] - List of demo-folders of selected Category



4) [P.2.OK] - Run runme.sh from selected demo-folders



5) [L] - List of demo-folders

C:\WINDOWS\system32\cmd.exe - bash x4pro2.sh

X4Pro-dialog

Run selected examples

[] 0	part1-0-sig	Retrieve and plot EXFOR cross sections (plain Python-program)
[*] 1	part1-1-sig	Retrieve and plot EXFOR cross sections (modular Python-program)
[*] 2	part1-2-da	EXFOR angular distributions
[] 3	part1-3-dap	EXFOR partial angular distributions
[] 4	part1-4-de	EXFOR energy spectra of outgoing particles
[] 5	part1-5-dae	EXFOR double differential cross sections
[] 6	part1-6-fy	EXFOR fission yield
[] 7	part1-7-covar	Retrieve and plot covariance data stored in EXFOR
[*] 8	part1-fortran	Retrieve data from EXFOR using SQL from Fortran programs
[] 9	part2-1-sig1	Retrieve and plot EXFOR/ENDF cross sections
[] a10	part2-2-da1an	EXFOR/ENDF angular distributions (Ei:fixed)
[] b11	part2-3-da1ei	EXFOR/ENDF angular distributions (angle:fixed)
[] c12	part2-4-de1eo	EXFOR/ENDF energy spectra of outgoing particles
[] d13	part2-5-dae1eo	EXFOR/ENDF double differential cross sections
[] e14	part2-6-fy	EXFOR/ENDF fission yield (mass distribution)
[] f15	part2-7-fye	EXFOR/ENDF fission yield (energy distribution)
[*] g16	part3-1-auto1	Automatically renormalize EXFOR CS using old monitor CS and new st
[*] h17	part3-2-user1	Correct data from EXFOR by user's program
[*] i18	part3-3-expert1	Apply experts corrections stored in X4Pro database
[*] j19	part3-4-auto2	Automatically renormalize EXFOR CS using old monitor CS and new st
[] k20	part3-5-ratio2sig	Retrieve ratios, calculate CS and uncertainties using latest stand
[] l21	part3-6-daleg2sig	Retrieve from EXFOR: Legendre coefficient L[0], calculate cross se
[*] m22	part3-7-legrs2da	EXFOR LEG/RS and SIG from the same Entry; calc. missing Legendre c
[*] n23	part3-8-leg2r33	Retrieve Legendre coefficient calculate angular distributions stor
[] o24	part5-0-sql	SQL to find zero values of PARITY, ERR-T and DATA-ERR
[] p25	part6-0-eval1flags	Generate summary tables with evaluators' flags and scores
[] q26	part6-1-eval2sig	Retrieve CS, ratios, monit-data; renormalize to new standards andS

100%

< OK >

<Cancel>

6) [S] - EXFOR database statistics

```

C:\WINDOWS\system32\cmd.exe - bash x4pro2.sh
---CHOICE=S
---EXFOR Statistics---
Database-file: ../x4sqlite1.db 2026-02-16,14:52:50 size=186M
Data in database:
--Database-uploaded: 2026-02-16,12:31:45
--Data-update: 2026-01-22
--Entry/Subent: 390/6692
--Datasets/points: 5380/251401
--wget-cache: 81/2026-02-16,14:52:49
--Datasets by Targets: (limit:8)
+-----+-----+
| Datasets | Target |
+-----+-----+
| 285      | O-16   |
| 233      | Al-27  |
| 162      | U-235  |
| 145      | Mn-55  |
| 127      | Co-59  |
| 122      | Ni-58  |
| 110      | C-12   |
| 109      | Fe-56  |
+-----+-----+
--Datasets by Incident particles: (limit:8)
+-----+-----+
| Datasets | Projectile | Help |
+-----+-----+
| 4846     | N          | Neutrons |
| 540      | P          | Protons  |
| 32       | HE3        | He-3     |
| 22       | A          | Alphas   |
| 17       | D          | Deuterons |
| 9        | He-6       |          |
| 4        | 0          | -        |
+-----+-----+
--Datasets by Quantity: (limit:8)
+-----+-----+
| Datasets | Quantity | Help |
+-----+-----+
| 3765     | CS       | Cross section data |
| 503      | DA       | Differential data with respect to angle |
| 251      | DAP      | Partial differential data with respect to angle |
| 170      | FY       | Fission product yields |
| 156      | RI       | Resonance integrals |
| 136      | DAE      | Differential data with respect to angle and energy |
| 133      | CSP      | Partial cross section data |
| 127      | POL      | Polarization data |
+-----+-----+
Press any key to continue...

```

7) [V] - Versions of platform and system components

```
C:\WINDOWS\system32\cmd.exe - bash x4pro2.sh

X4Pro

-----+
| X4Pro: fully relational EXFOR database. |
| Professional Edition.                  |
| Version 2026-02-23.                   |
| Demo-examples on Python/SQL/Fortran:  |
| search/retrieve/plot/recalculate data. |
| v.zerkin@gmail.com, IAEA-NRDC, 2021-2026 |
|-----+
Run:          2026-02-23,14:53:41
Platform:     MINGW32_NT-6.2 1.0.19(0.48/3/2) i686
Computer:     ZVV-MINI-PC
Shell:        GNU bash, version 3.1.23(1)-release (i686-pc-msys)
Bash-ver:     3.1.23(1)-release
Script:       x4pro2.sh
Now Dir:      /g/projects/zerkin/pdev/x4pro7-dist/x4pro14
---Versions---
1) python:    [3.9.5]
2) pip:       [21.1.2]
3) matplotlib: [3.5.2]
4) plotly:    [5.3.1]
5) requests:  [2.26.0]
6) pandas:    [1.3.3]
7) kaleido:   [0.2.1]
8) couchdb:   [1.2]
9) gfortran:  GNU Fortran (MinGW.org GCC Build-2) 9.2.0
10) sqlite3:  3.47.2 2024-12-07 20:39:59 (64-bit)
11) dialog:   Version: 1.1-20100119
12) database-file: ../x4sqlite1.db date:2026-02-16 size=186M
    database-data: 2026-02-16,12:31:45 Entry:390 Datapoints:251401

Press any key to continue...
```